

This document serves as a brief record of time spent, compute used, AI tools used and unfinished work.

The assignment was completed in around 3 hours with the first 2 hours were spent on ideation and trials while communicating with LLM's like Claude and ChatGpt. The time for the final running of code on the compute of a single i5, RTX 4050 laptop was 13 min 27 sec. The final portion of the time was spent on formatting the directory, generating the required report and writing this document. As mentioned above, AI LLM's were used for generating the actual files as well as formatting of both the directory and the report. The actual implementation and this document are done without any AI assistance.

Some Unfinished work includes:

- No hyperparameter tuning on the GBM/ridge models
- site_ridge/E_ridge were only run on the block/domain splits where the failure showed up and wasn't re-run on random/posgroup for completeness.
- No formal significance test comparing M1 vs. M2 vs. E beyond the bootstrap CIs already reported

Almost all of these were due to time constraints.